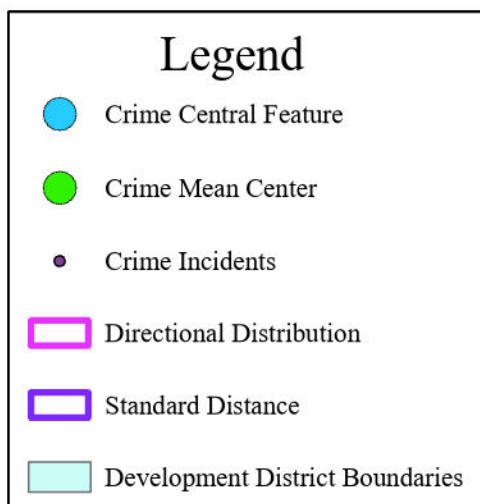
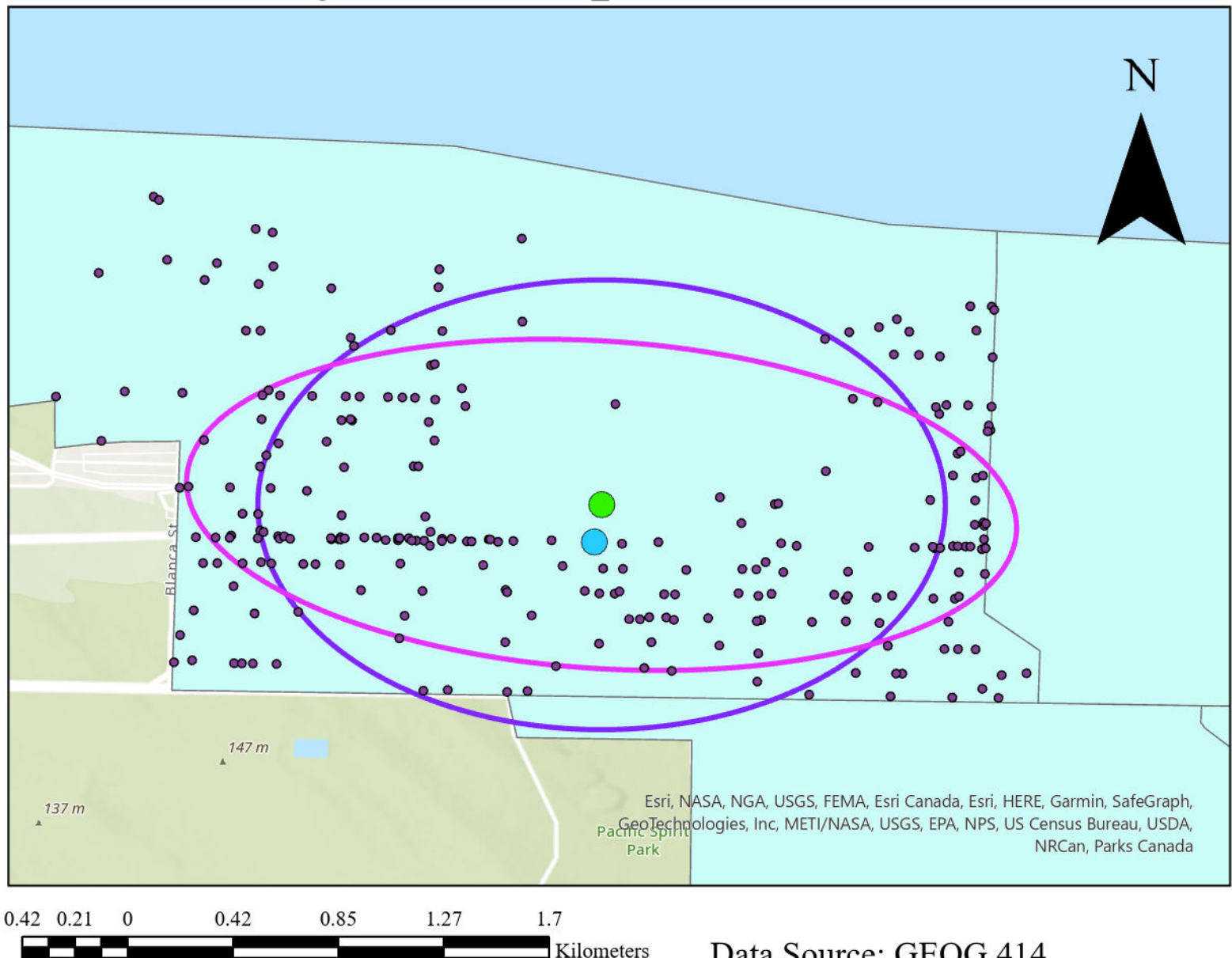


2009 Vancouver Crime in West Point Grey Development District

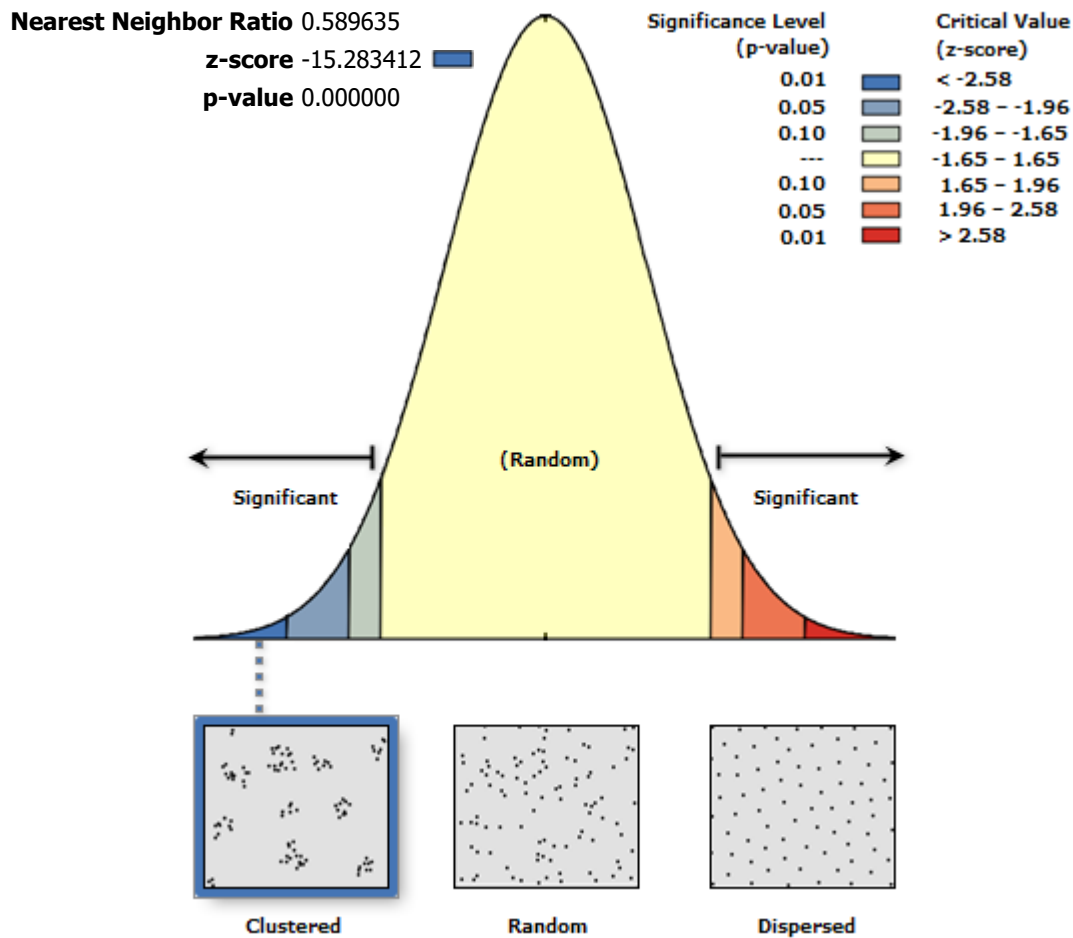


Data Source: GEOG 414
Projected Coordinate System: WGS
1984 UTM Zone 10

Author: Grady Zappone
Date: 19 November 2023

A map displaying the distribution and centrographic statistics of crime incidents in the West Point Grey District of Vancouver in 2009.

Average Nearest Neighbor Summary



Given the z-score of -15.283412, there is a less than 1% likelihood that this clustered pattern could be the result of random chance.

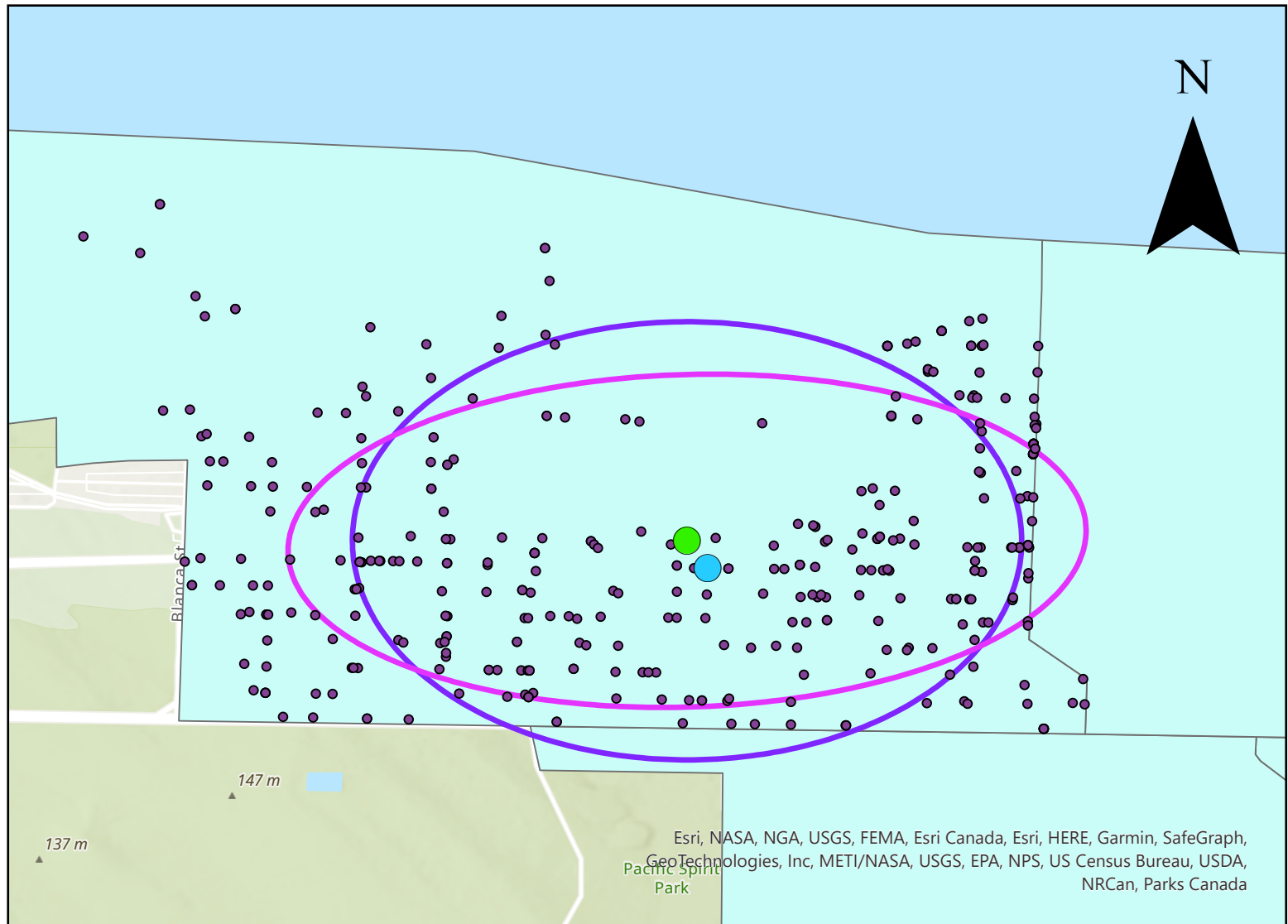
Average Nearest Neighbor Summary

Observed Mean Distance	34.8781 Meters
Expected Mean Distance	59.1520 Meters
Nearest Neighbor Ratio	0.589635
z-score	-15.283412
p-value	0.000000

Dataset Information

Input Feature Class:	VancouverCrime_2009
Distance Method:	MANHATTAN
Study Area:	5304418.078730
Selection Set:	False

2019 Vancouver Crime in West Point Grey Development District



0.42 0.21 0 0.42 0.85 1.27 1.7
Kilometers

Legend

-  Crime Central Feature
-  Crime Mean Center
-  Crime Incidents
-  Directional Distribution
-  Standard Distance
-  Development District Boundaries

Data Source: GEOG 414

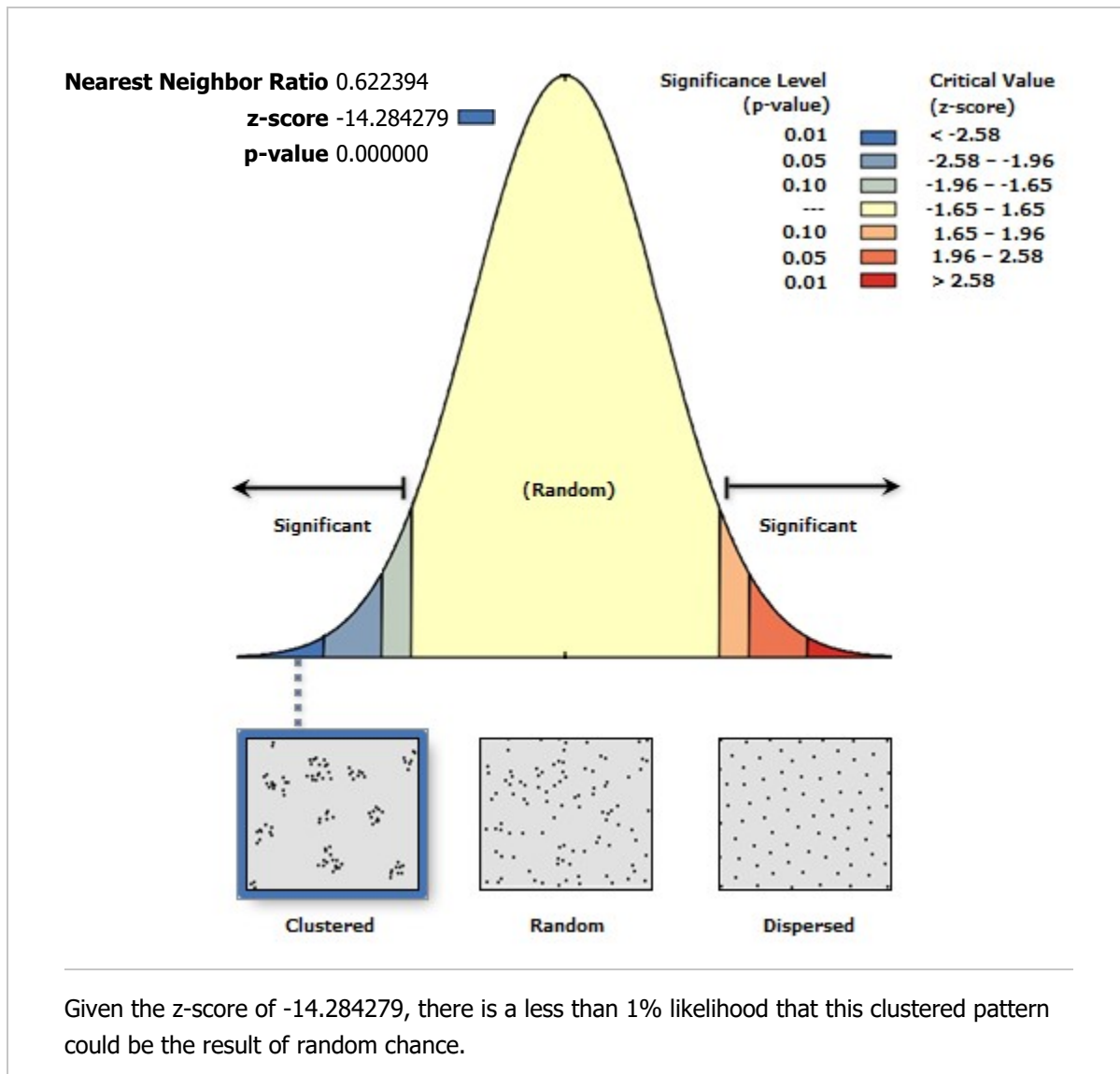
Projected Coordinate System: WGS
1984 UTM Zone 10

Author: Grady Zappone

Date: 19 November 2023

A map displaying the distribution and
centrographic statistics of crime
incidents in the West Point Grey
District of Vancouver in 2019.

Average Nearest Neighbor Summary



Average Nearest Neighbor Summary

Observed Mean Distance	36.2465 Meters
Expected Mean Distance	58.2372 Meters
Nearest Neighbor Ratio	0.622394
z-score	-14.284279
p-value	0.000000

Dataset Information

Input Feature Class:	VancouverCrime_2019

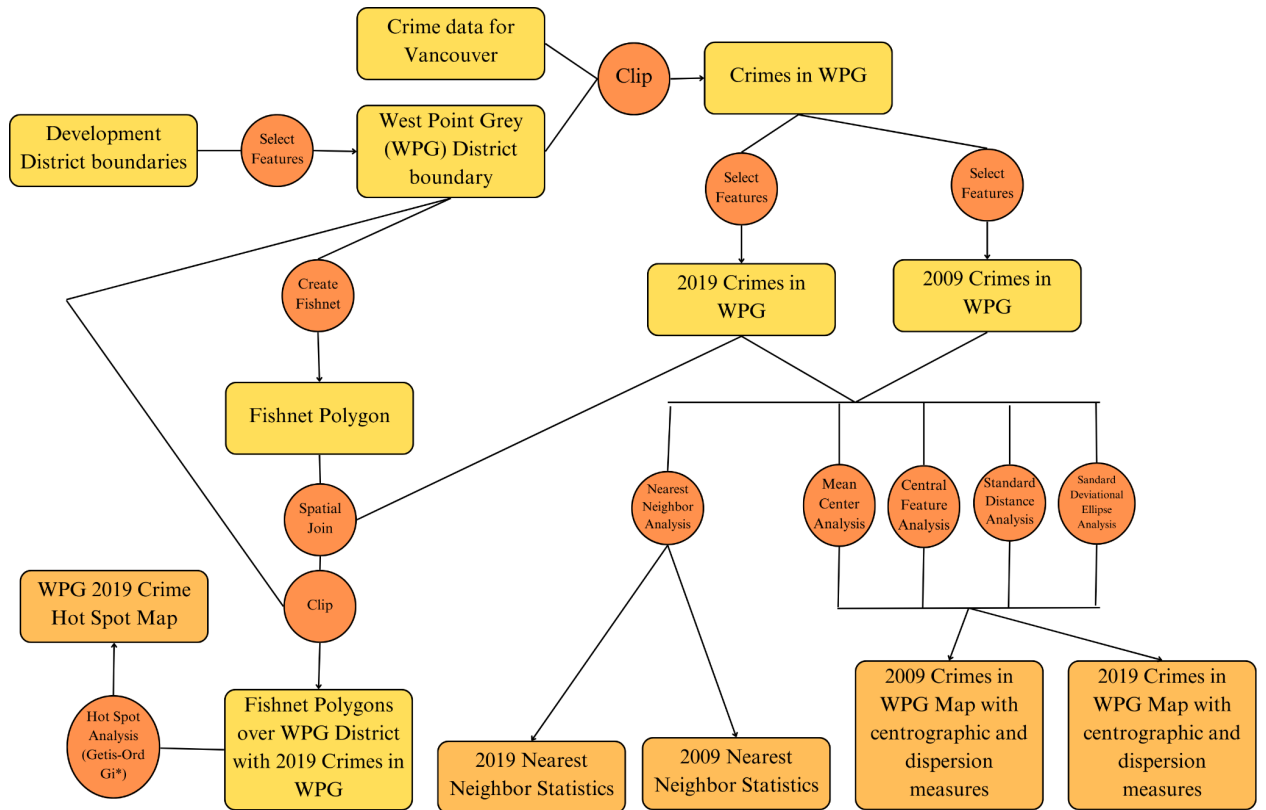
Distance Method:	MANHATTAN
Study Area:	5304418.078730
Selection Set:	False

Lab 4 Analysis

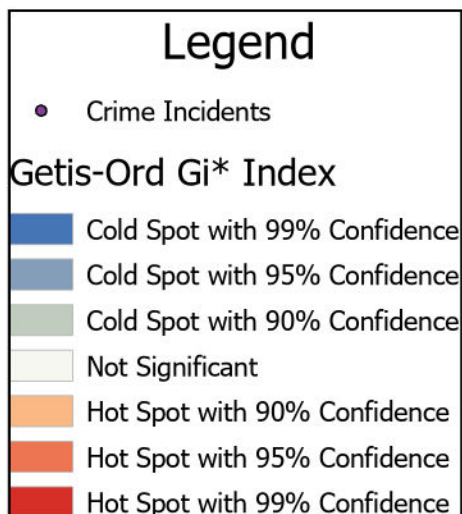
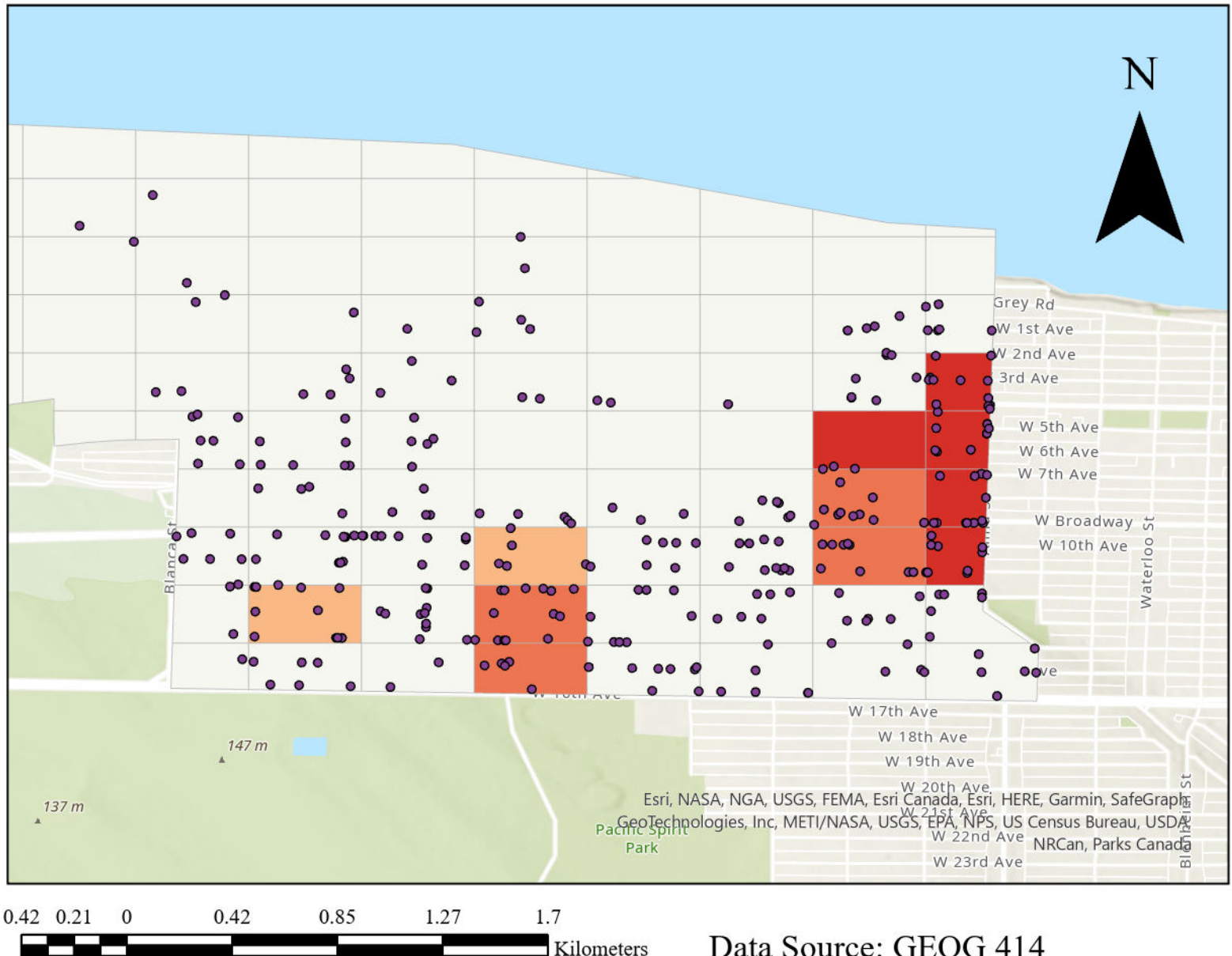
The maps and average nearest neighbor results can be used to determine whether the clustering patterns of auto thefts have changed significantly between 2009 and 2019 in the West Point Grey Development District of Vancouver. The maps show the distribution of crime incidents, with a mean center, central feature, standard deviational ellipse and a standard distance displayed. Using these features, it is clear that the center of crime incidents has shifted slightly to the East between 2009 and 2019. Additionally, the modest contraction in the directional distribution ellipse along the major axis (East-West) shows the average range of incidents has shrunk, meaning the crimes are more clustered in the East than in 2009. The standard deviation ellipse depicts a similar trend by also having a smaller radius and its centroid moving slightly to the East.

Next, we can analyze the average nearest neighbor statistics: The nearest neighbor ratio is a measure of clustering, with a value less than 1 indicating a clustered pattern, and a value greater than 1 indicating a dispersed pattern. For 2009, the observed mean distance between nearest neighbors was 34.8781 meters and the expected mean distance was 59.1520 meters. This means the distribution of crimes were clustered since the actual distance was less than the expected distance. The nearest neighbor ratio was 0.589635, with a p-value very close to 0, meaning there is almost no chance this clustering would happen randomly. For 2019, the nearest neighbor ratio was 0.622394. Both years show a nearest neighbor ratio of less than 1, indicating a clustered pattern, with extremely low p-values suggesting that this pattern is very unlikely to be due to random chance. However, the larger NNR in 2009 suggests the clustering of crime incidents was stronger in 2009 than in 2019. In comparing the observed mean distances, we see a slight increase from 2009 to 2019, suggesting that the incidents are slightly more spread out in 2019. Despite the slight change in the clustering of crimes, both distributions were significantly clustered in both years. The slight decrease in clustering intensity could be due to various factors, such as changes in urban development or policing strategies.

Cartographic Model



2019 Vancouver Crime Hot Spots in West Point Grey Development District



Data Source: GEOG 414
Projected Coordinate System: WGS 1984 UTM Zone 10

Author: Grady Zappone
Date: 19 November 2023

A map displaying the hot and cold spots in the 2019 West Point Grey District, Vancouver crime incidents according to an Getis-Ord Gi* analysis.